Memo Monday

Use a pen to complete the test. Show all your steps and simplify your answers. Use interval notation, where applicable. You are not allowed to use a calculator. The test is printed on both pages.

Let $f(x) = x^3 - 3x^2 - 9x + 1$. Given that $f'(x) = 3x^2 - 6x - 9$, and $f''(x) = 6x - 6$.

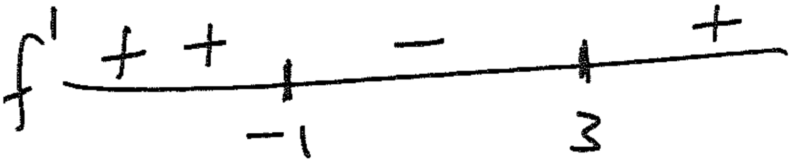
QUESTION 1 [1 mark]
Determine the critical numbers for f.

$$f'(x)=0 \Rightarrow 3(x^2-2x-3)=0$$
$$\Rightarrow 3(x-3)(x+1)=0$$
$$\Rightarrow x=3 \text{ or } x=-1$$

Critical numbers: $x=-1$ and $x=3$

QUESTION 2 [2 marks]
Use the first derivative test to find the local extremes of f.

$$f'(x)=3(x-3)(x+1)$$



$f'(-10)>0$
$f'(0)<0$
$f'(10)>0$

Since f increases on $(-\infty,-1)$ and decreases on $(-1,3)$, it has a local max of $f(-1)=(-1)^3-3+9+1$
$= 6$.

Local min value is $f(3)=(3)^3-27-9(3)+1$
$=-26$

QUESTION 3 [2 marks]
Determine the concavity of f.

$f''(x) = 6(x-1)$

$f'' > 0 \Rightarrow x-1 > 0 \Rightarrow x > 1$

$\Rightarrow$ f is concave up on $(1, \infty)$

f is concave down on $(-\infty, 1)$

QUESTION 4 [1 marks]
Find the inflection points of f.

Concavity changes at $x=1$ $\Rightarrow$ Inflection point at $x=1$.

$f(1) = 1-3-9+1 = -10$

$\Rightarrow$ Inflection point is $(1,-10)$.

$f(x) = x^3 - 3x^2 - 9x + 1$

QUESTION 5 [2 marks]
Find the absolute extreme values of f on the interval [-2,0].

Critical numbers in $[-2,0]$: $x=-1$

$f(-2) = (-2)^3 - 3(4) + 18 + 1 = -1$
$f(-1) = -1-3+9+1 = 6$
$f(0) = 1$

Abs. max of f on $[-2,0]$ is 6

Abs. min of f on $[-2,0]$ is -1.

Memo Tuesday, Thursday

Use a pen to complete the test. Show all your steps and simplify your answers. Use interval notation, where applicable. You are not allowed to use a calculator.

Let $f(x) = x^2e^{-x}$. Given that $f'(x) = e^{-x}(2x - x^2)$, and $f''(x) = e^{-x}(x^2 - 4x + 2)$.

QUESTION 1 [1 mark]

Determine the critical numbers for f.

$f'(x) = xe^{-x}(2-x) = 0 \Rightarrow x=0$ & $x=2$

$\Rightarrow$ Critical numbers: $x=0$ and $x=2$.

QUESTION 2 [2 marks]

Use the first derivative test to find the local extremes of f.

$f'(x) = \dfrac{x(2-x)}{e^x}$

$f'(-10) < 0$

$f'(1) > 0$

$f'(10) < 0$

f': $-$ | $+$ | $-$

at 0 and 2

$\Rightarrow f$ decreases on $(-\infty, 0)$ and increases on $(0,2)$

$\Rightarrow$ local min at $x=0$

Local min. value is $f(0) = 0$

f increases on $(0,2)$ and decreases on $(2,\infty)$

$\Rightarrow$ local max of $f(2) = \dfrac{4}{e^2}$ at $x=2$.

QUESTION 3 [2 marks]
Determine the concavity of f.

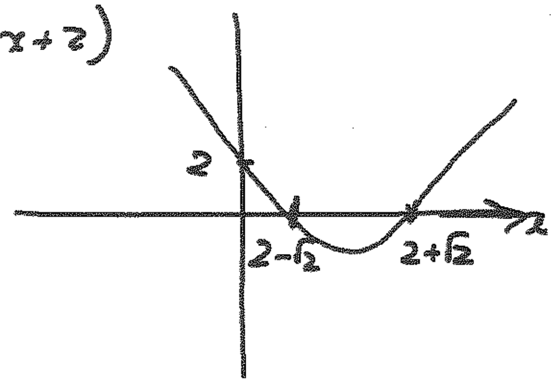
$$f''(x) = e^{-x}(x^2-4x+2)$$

$$f''(x) > 0 \Rightarrow x^2-4x+2 > 0$$

If $x^2-4x+2=0$

$$\Rightarrow x = \frac{4 \pm \sqrt{16-8}}{2}$$

$$= \frac{4 \pm \sqrt{8}}{2} = \frac{4 \pm 2\sqrt{2}}{2} = 2 \pm \sqrt{2}$$



f is concave up on $(-\infty, 2-\sqrt{2}) \cup (2+\sqrt{2}, \infty)$

f is concave down on $(2-\sqrt{2}, 2+\sqrt{2})$

QUESTION 4 [1 mark]
Find the inflection points of f. Do not simplify the function value.

f changes concavity at $x = 2-\sqrt{2}$ and $x = 2+\sqrt{2}$

Inflection point:

$$(2-\sqrt{2}, f(2-\sqrt{2})) \text{ and } (2+\sqrt{2}, f(2+\sqrt{2}))$$

QUESTION 5 [2 marks]
Find the absolute extreme values of f on the interval [0,3].

$$f(x) = x^2 e^{-x}$$

One critical value in $(0,3)$, namely $x=2$

$f(0) = 0$
$f(2) = 4/e^2$
$f(3) = 9/e^3$

Absolute min on $[0,3]$ is $f(0)=0$
Absolute max on $[0,3]$ is $f(2) = 4/e^2$

Memo Friday

Use a pen to complete the test. Show all your steps and simplify your answers. Use interval notation, where applicable. You are not allowed to use a calculator.

Let $f(x) = xe^{-2x}$. Given that $f'(x) = e^{-2x}(1-2x)$, and $f''(x) = 4e^{-2x}(x-1)$

QUESTION 1 [1 mark]
Determine the critical numbers for f.

$f'(x) = e^{-2x}(1-2x) = 0 \Rightarrow 1-2x=0 \Rightarrow x = \frac{1}{2}$

$\Rightarrow x = \frac{1}{2}$ is the critical number of f.

QUESTION 2 [2 marks]
Use the first derivative test to find the local extremes of f.

f': $+$ | $\frac{1}{2}$ | $-$

$f'(0) = 1 > 0$

$f'(1) = e^{-2}(1-2) < 0$

$\Rightarrow f$ increases on $(-\infty, \frac{1}{2})$ and decreases on $(\frac{1}{2}, \infty)$

$\Rightarrow$ local max is $f(\frac{1}{2}) = \frac{1}{2}e^{-1} = \frac{1}{2e}$.

$f(x) = xe^{-2x}$

QUESTION 3 [2 marks]
Determine the concavity of f.

$f''(x) = 4e^{-2x}(x-1)$

$f'' > 0 \Rightarrow x-1 > 0 \Rightarrow x > 1$

$\Rightarrow f$ is concave up on $(1,\infty)$

and f is concave down on $(-\infty, 1)$

$f(x) = xe^{-2x}$

QUESTION 4 [1 mark]
Find the inflection points of f.

f changes concavity at $x = 1$

$\Rightarrow$ Inflection point $(1, f(1))$

$= (1, \frac{1}{e^2})$

$f(x) = xe^{-2x}$
$f(1) = 1e^{-2} = \frac{1}{e^2}$

$f(x) = xe^{-2x}$

QUESTION 5 [2 marks]
Find the absolute extreme values of f on the interval [0,1].

$f(0) = 0$
$f(\frac{1}{2}) = \frac{1}{2}e^{-1} = \frac{1}{2e}$
$f(1) = 1e^{-2} = \frac{1}{e^2}$

$\Rightarrow$ Absolute max is $\frac{1}{2e}$ (at $x = \frac{1}{2}$).

Absolute min is 0 (at $x = 0$).